

NAME: _____ CLASS: _____

MZUZU DIOCESE
2021 MALAWI SCHOOL CERTIFICATE OF EDUCATION
EXAMINATION
BIOLOGY

Tuesday, 3rd August

Subject Number: M022/1
Time Allowed: 2hours

PAPER I
(100 marks)

Instructions

1. **This paper contains 13 pages. Please check.**

2. Before beginning write your **name** at the top of the question paper and on all other sheets.

3. This paper contains sections **A** and **B**.
Answer all the 14 questions in both sections.
Some can be answered quickly, but others require considerable thought and may take longer.

4. Write your answers on the question paper in the spaces provided. The maximum number of marks for each answer is indicated against each question

5. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
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12			
13			
14			

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SECTION A (70 marks)

Answer all the questions in this section

1. Some washing powders are described as having ‘biological’ action because they contain enzymes. These powders are particularly useful for removing stains of blood, eggs, chocolate etc.

a. Explain why biological washing powders are;

(i) less effective if the water used for washing clothes is cold.

(1 mark)

(ii) more effective if the water used for washing is luke-warm.

(2 marks)

b. Name **two** examples of enzymes that are contained in such biological washing powders.

(i) _____

(ii) _____
(2 marks)

c. Suggest **one** reason why biological washing powders are more economical than ordinary washing powders to remove stains.

(2 marks)

2. a. Explain why patient’s blood and the dialysis fluid move in opposite direction in the dialysis machine.

(2 marks)

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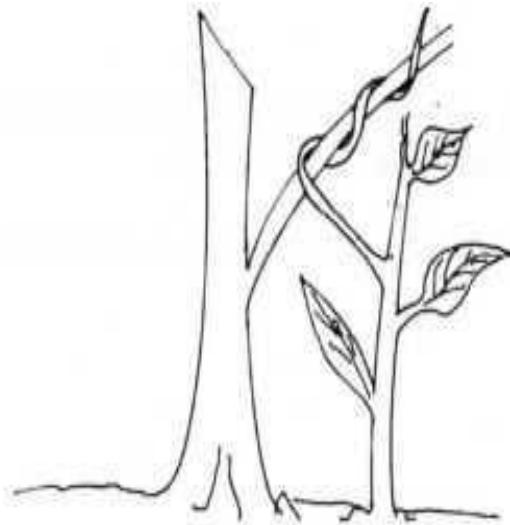
- b. Briefly explain how Ant- Diuretic Hormone (ADH) influences the amount and concentration of urine

(4 marks)

- c. Why are people with kidney failure advised to reduce salt intake in their diet.

(2 marks)

3. The diagram below illustrates a tropic response by a certain plant. Use it to answer questions that follow.



- a. Name the type of response

(1 mark)

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b. Explain how the response illustrated above occurs

(3 marks)

c. What is the importance of this type of response?

(1 mark)

d. Give **two** examples of plants that exhibit nastic responses.

(2 marks)

4. **Figure 1** shows a confused scientist looking at a big-eyed mouse in a cage whose genotype is either homozygous dominant or heterozygous.



Figure 1

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- a. Name the phenotype that is being studied in this investigation.

(1 mark)

- b. Why is it difficult to identify the genotype of the mouse by using its phenotype?

(1 mark)

- c. Briefly describe what the scientist can do in order to establish the genotype of the mouse in the cage.

(4 marks)

5. **Figure 2** is a diagram showing some of the processes taking place in living plant cells at different times of the day.

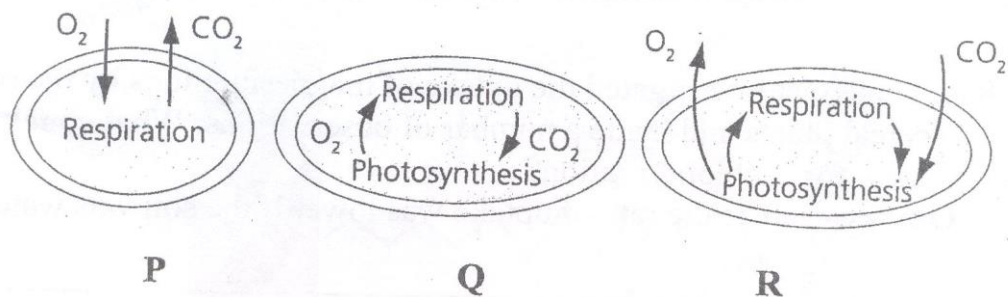


Figure 2

- a. (i) Which diagram shows that the plant cell is exposed to day light?

(1 mark)

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(ii) Give a reason for your answer.

(1 mark)

b. Describe what is happening in the cell at Q?

(2 marks)

6.a. In the locomotion of birds, what do the following terms mean?

(i) Upstroke

(1 mark)

(ii) Downstroke

(1 mark)

b. Explain why recovery stroke is important.

(2 marks)

7. **Figure 3** is a diagram showing two invertebrate animals. Use it to answer the questions that follow.

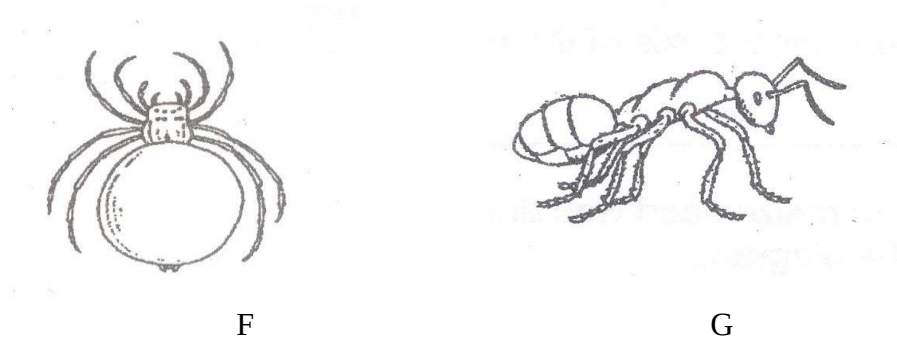


Figure 3

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a. To which group of arthropods does each of these organisms belong?

F _____

G _____

(2 marks)

b. State any **two** features that are unique to organisms of group 'F' named above.

i _____

ii _____

(2 marks)

c. Mention **one** observable feature which make the two organisms belong to arthropods.

_____ (1 mark)

8. **Figure 4** is a diagram showing part of placenta. Use it to answer the questions that follow.

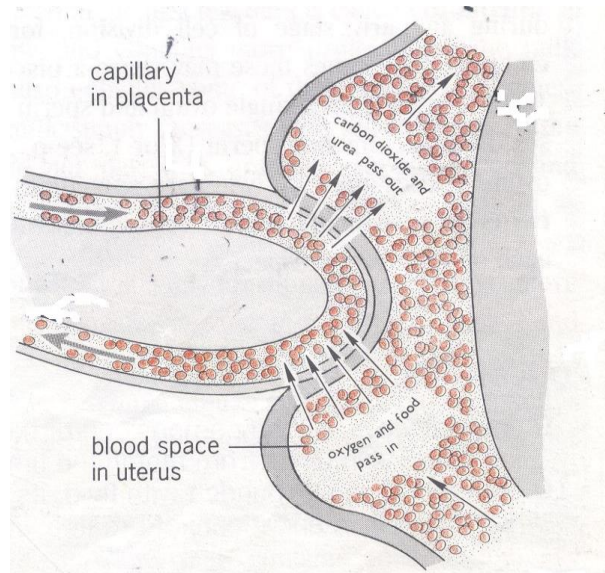


Figure 4

a. State **two** functions of placenta that are shown in the diagram above.

(i) _____

(ii) _____

(2 marks)

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- b. Explain how the placenta acts as;
(i) an excretory system for the foetus.

(1 mark)

- (ii) an ileum for the foetus.

(1 mark)

- iii. Mention **two** organs in the baby which do not function when it is in the mother's womb.

(i) _____

(ii) _____

(2 marks)

- c. Explain the roles of any **two** hormones in the menstrual cycle.

(4 marks)

- 9 a. Define coordination

(1 mark)

- b. State any **two** differences between nervous system and endocrine system.

(2 marks)

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c. Mention the **three** main steps involved in conditioning organisms.

(3 marks)

10. **Figure 5** shows one of the nutrient cycles in nature. Use it to answer the questions that follow.

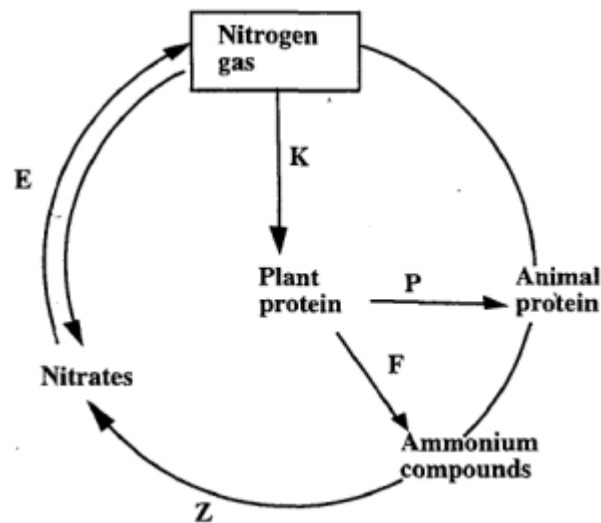


Figure 5

(i) Name the processes taking place at **F** and **Z** .

F: _____

Z: _____

(2 marks)

(ii) Explain how the process at **K** takes place.

(2 marks)

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(iii) Under what conditions does the process at **E** take place?

(1 mark)

b. Describe any **two** groups of plants according to the ecosystems they are adapted to.

(4 marks)

11. a. Explain why large animals need a circulatory system

(2 marks)

b. Explain how lymph is formed in the human body.

(2 marks)

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SECTION B (30 marks)

Answer all the questions in this section.

12. Explain how natural and artificial immunities are acquired in the human body. Your answer should be in an essay form.

[illegible]

(10 marks)

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13. Explain any **five** human activities that cause environmental degradation. Your answer should be in an essay form.

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(10 marks)

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14. Explain any **five** ways how biotechnology can be used to improve quality of human life. Your answer should be in an essay form.

[illegible]

(10 marks)

END OF QUESTION PAPER